

lot information definitions.⁹⁶¹ The Commission explained that the proposed compliance date would significantly move up the date by which round lot and odd-lot information would be more widely available in the national market system.⁹⁶²

Several commenters raised concerns about the proposed compliance date, stating that 90 days was not enough time to implement the round lot and odd-lot information definitions.⁹⁶³ In stating that a longer timeframe was needed, some commenters stated their views of the challenges entailed in implementing the changes.⁹⁶⁴ One commenter stated, “[t]he technical and operational requirements to implement the definition changes will necessitate distinct product changes in the systems of literally hundreds of exchanges, vendors, and subscribers, each with different development priorities and system capabilities.”⁹⁶⁵ The commenter stated that 90 days would not be enough time for the exclusive SIPs, data vendors and subscribers to accommodate the changes, and cautioned that “hastily made changes or missed delivery dates could result in not just a failure to provide odd-lot quotation data but also disrupt the flow of other core data to the market.”⁹⁶⁶ The commenter urged the Commission not to adopt the 90-day compliance timeframe and instead, after adoption of the Proposing Release, allow time for industry

⁹⁶¹ See Proposing Release, supra note 11, at 80300; see id. at n.399 and accompanying text.

⁹⁶² See Proposing Release, supra note 11, at 80298.

⁹⁶³ See, e.g., NYSE Letter I at 7-8; CTA, CQ, UTP Plans Operating Committees Letter at 3; MEMX Letter at 7; FISD Letter at 2, 3, 4; Nasdaq Letter I at 3; Cboe Letter II at 11; FIF Letter at 14; Cboe Letter III at 10 n.18. See also Fidelity Letter at 16.

⁹⁶⁴ See, e.g., NYSE Letter I at 7-8; CTA, CQ, UTP Plans Operating Committees Letter at 2, 3; FISD Letter at 2, 3, 4; FIF Letter at 14; Cboe Letter II at 11.

⁹⁶⁵ See FISD Letter at 3.

⁹⁶⁶ Id.

consultation to develop an implementation plan.⁹⁶⁷ Two commenters, while supporting the odd-lot information definition adopted in the MDI Rules, recommended only implementing the BOLO due to the complexity and time it would take to implement the odd-lot information definition by many market participants.⁹⁶⁸

Three commenters suggested an implementation timeframe of at least one year.⁹⁶⁹ One commenter explained that the changes to the round lot definition would require programming changes by the exclusive SIPs and the market participants that receive SIP data, as well as testing of the changes at the exchanges, exclusive SIPs and customer levels,⁹⁷⁰ and that it would likely take longer than one year for the exclusive SIPs and exchanges to implement the proposed odd-lot changes.⁹⁷¹ The Operating Committees for the Equity Data Plans stated that the implementation timeframe for the exclusive SIPs would likely extend beyond one year due to, among other things, “the time needed for system design, [to] procure necessary equipment, and accommodate industry testing.”⁹⁷² Another commenter stated that the proposed 90-day timeframe was too aggressive, did not consider “technical realities,” and suggested an implementation period of at least one year.⁹⁷³

⁹⁶⁷ See FISD Letter at 4. The commenter stated that the Operating Committees of the Equity Data Plans considered a 10-12 month implementation process in their proposal to add odd-lot data to the exclusive SIP feeds. Id.

⁹⁶⁸ See BlackRock Letter at 12. See also MEMX Letter at 7.

⁹⁶⁹ See NYSE Letter I at 8; CTA, CQ, UTP Plans Operating Committees Letter at 3; Nasdaq Letter I at 3.

⁹⁷⁰ See NYSE Letter I at 7.

⁹⁷¹ See NYSE Letter I at 8.

⁹⁷² CTA/CQ/UTP Plans Operating Committees Letter at 2, 3.

⁹⁷³ See Nasdaq Letter I at 3.

In light of the comments, the Commission is modifying the compliance date for the round lot and odd-lot information definitions. For implementation of the round lot definition⁹⁷⁴ and the round lot indicator,⁹⁷⁵ the compliance date will be the first business day of November 2025. The Commission calculated this deadline based on two main factors. First, the compliance date is approximately 12 months after the effective date, which is consistent with what commenters suggested was necessary for systems changes and testing. Second, the compliance date provides sufficient time for any exchanges that have defined round lots in their rules to file proposed rule changes pursuant to section 19(b) of the Exchange Act⁹⁷⁶ and rule 19b-4⁹⁷⁷ thereunder to reflect the new round lot definition.⁹⁷⁸

The compliance date for the odd-lot information definition⁹⁷⁹ and Rule 603(b)(3) (with respect to the requirement that every national securities exchange on which an NMS stock is traded and national securities association must make available to the exclusive SIPs all data necessary to generate odd-lot information, and the collection, consolidation and dissemination of odd-lot information by the exclusive SIPs) will be the first business day of May 2026, which is approximately 18 months after the effective date of the Adopting Release. The Commission is providing a modified compliance date for the odd-lot information definition, consistent with what industry comment suggested was necessary for technical and operational requirements,⁹⁸⁰ due to several factors. First, the exclusive SIPs will likely have to make more changes to their

⁹⁷⁴ Rule 600(b)(93).

⁹⁷⁵ Rule 600(b)(89)(i)(E) and Rule 600(b)(89)(iv).

⁹⁷⁶ 15 U.S.C. 78s(b).

⁹⁷⁷ 17 CFR 240.19b-4.

⁹⁷⁸ See Proposing Release, supra note 11, at 80300 n.408.

⁹⁷⁹ Rule 600(b)(69).

⁹⁸⁰ See supra notes 969 and 973 (suggesting an implementation process of approximately one year).

systems to accommodate the odd-lot information definition than to implement the round lot definition. Specifically, the exclusive SIPs will need to collect more data, consolidate it, and disseminate it as odd-lot information. In addition, the exclusive SIPs will need to calculate and disseminate the BOLO. The Commission continues to believe that both the changes to the odd-lot information definition and the dissemination of the BOLO are independently important, and the additional time allotted to comply with the odd-lot information definition addresses the concerns from commenters regarding the complexity or operational risks that may arise with making odd-lot information changes in a compressed timeline. Second, the effective national market system plan(s) may also need to assess whether plan amendments will be necessary to conform such plans to the odd-lot information definition, and to file any such amendments with the Commission pursuant to rule 608. Finally, market participants may need to update their systems that accept SIP data to reflect odd-lot information.

Accordingly, extending the compliance deadlines for the implementation of the round lot and odd-lot information definitions will address the concerns raised by commenters and provide additional time for market participants to make the changes necessary to implement the definitions.

VII. Economic Analysis

A. Introduction

The most common method of trading in NMS stocks by registered exchanges today is the limit order book matching system, a mechanism that securities exchanges use to bring together orders of multiple buyers and sellers of securities and have those orders interact. It acts as a central hub where participants' priced buy and sell orders can be ranked, displayed, and matched based on programmed

rules established by the providing registered exchange. As such, the limit order book matching system facilitates efficient and competitive markets.⁹⁸¹

Imagine an order book in which a buyer's or seller's order could be displayed at any pricing increment, no matter how small. In this scenario, assume a liquidity provider wants to buy a stock. The provider sees the book with the prices at which others are willing to buy. Because in this hypothetical market there are no restrictions on an entry price point, the liquidity provider can jump ahead of those other providers by offering to buy at a price that is infinitesimally higher. This is what is known as "pennying."⁹⁸² The problem with pennying is that it creates a disincentive for liquidity providers to post buy or sell orders, because they know that a second trader can step ahead with an infinitesimally better price. This leads to lower priced offers to buy and higher priced offers to sell – namely a wider quoted bid-ask spread.

Recognizing this market failure, the Commission in 2005 adopted⁹⁸³ a market-wide requirement that venues could not display, rank, or accept orders in increments less than a penny.⁹⁸⁴ The 2005 adoption of Rule 612 limited the scope of pennying, but it did so at the inevitable cost of introducing a floor, namely one cent, below which the quoted bid-ask spread could not fall.

⁹⁸¹ See Anthony Clarke, Demystifying the Central Limit Order Book (CLOB): Everything You Need to Know (Apr. 21, 2023), available at <https://www.nasdaq.com/articles/demystifying-the-central-limit-order-book-clob-everything-you-need-to-know>.

⁹⁸² See supra section I.A.1.

⁹⁸³ See generally, Regulation NMS Adopting Release, supra note 4.

⁹⁸⁴ Specifically, preexisting Rule 612 of Regulation NMS prohibited a national securities exchange, national securities association, ATS, vendor, or broker or dealer from displaying, ranking, or accepting quotations, orders, or indications of interest in any NMS stock priced in an increment smaller than \$0.01 if the quotation, order, or indication of interest is priced equal to or greater than \$1.00 per share. If the quotation, order, or indication of interest is priced less than \$1.00 per share, the minimum pricing increment is \$0.0001.

Though a minimum tick is necessary, placing a floor on the spread introduces distortions into the market. The price of liquidity will be artificially high for some stocks, leading to a surplus, similar to a goods market for which prices were artificially high. This creates rents which accrue to some market participants at the expense of others. By reducing the minimum pricing increment for a defined subset of stocks, the adopted amendments to Rule 612 free the price of liquidity from its current constraint, allowing it to approach its natural level. At the same time, as described in greater detail below, the adopted amendments maintain a minimum (but smaller) pricing increment necessary for the proper functioning of financial markets' limit order books.

Freeing the spread from the binding constraint of one penny will bring a number of benefits, including lower transaction costs. For some stocks currently constrained at a penny, the spread will, under the amended rules, at times be a half-penny, a substantial reduction in the quoted price of accessing liquidity. This reduction, while beneficial, brings into the spotlight the cap on the access fee, which has been 0.30 cents. Absent a reduction in the maximum access fee, a round-trip buy and sell for stocks quoted at the new half-penny tick would require paying more in fees (0.60 cents) than in the spread itself (0.50 cents).

The practice of charging at or near the access fee cap has persisted over time. Regulation NMS establishes the NBBO. Because the NBBO is protected,⁹⁸⁵ many exchanges charge the maximum amount allowed to access the quote. This allows the exchange to subsidize liquidity providers with a rebate, reducing spreads (to acquire more volume, due to traders' need to access the protected quote). While the quoted spread may be lower, the cost to investors is not; this is

⁹⁸⁵ See supra note 42 and accompanying text discussing and defining protected quotes.

because gains from the lower spread are counteracted by the access fee. On the other hand, the high access fee and rebate can lead to a loss of price coherence when the spread is less than twice the fee. For stocks that remain tick constrained, as some may, the rebate distorts the supply of liquidity. Finally, fees and rebates that are high as a percentage of the quoted spread introduce complexity, and potential conflicts of interest. Lowering the access fee to 10 mils restores price coherence and alleviates these costs.

The Commission is also requiring that these fees and rebates be determinable at the time of trade execution. Opacity and complexities in current exchange fees and rebates make these more distortive than otherwise.⁹⁸⁶ With new Rule 610(d), the Commission is taking an incremental step in ameliorating the opacity in fees and rebates, reducing information asymmetries and lessening the potential for agency conflict between brokers and their customers.

Finally, the Commission has accelerated the implementation of the round lot, and odd-lot information definitions while providing more time for the necessary systems changes to implement the definitional changes than what was proposed. These amendments will allow the benefits of these rules to accrue to market participants in a timely manner.

Below, we explain why these amendments increase efficiency and competition and bring benefits that will accrue to the broad range of participants in U.S. equity markets. We also discuss the costs of these amendments. The Commission has considered the economic effects of the amendments and, wherever possible, the Commission has quantified the likely economic

⁹⁸⁶ As discussed in sections VII.D.2, VII.D.3, and VII.E.1, fees and rebates create a potential conflict for a broker in situations where transaction fees, which are paid by the broker, potentially conflict with execution quality, which is incurred by the customer. This conflict, if acted on, can lead to inefficient order routing and worse transaction outcomes for customers; it can also lead to an inefficient incorporation of information into stock prices, harming market efficiency.

effects of the amendments.⁹⁸⁷ The Commission is providing both a qualitative assessment and quantified estimates of the potential economic effects of the amendments where feasible. The Commission incorporated data and other information to assist it in the analysis of the economic effects of the amendments. However, as explained in more detail below, the Commission is unable to quantify certain economic effects because the Commission does not have, and in certain cases cannot reasonably obtain, data that may inform the Commission on certain economic effects. Further, even in cases where the Commission has data, it is not practicable to quantify certain economic effects due to the number and type of assumptions necessary, which render any such quantification unreliable. Our inability to quantify certain costs, benefits, and effects does not imply that such costs, benefits, or effects are less significant.

B. Broad Economic Considerations

1. Liquidity and Spread

A key component of market liquidity is the limit order book. Liquidity providers submit limit orders to buy (“bid”) and sell (“ask”) stock at specified prices and quantities. Liquidity demanders trade against these limit orders. The quoted (bid-ask) spread for a stock is the difference between the lowest displayed ask price and the highest displayed bid price.⁹⁸⁸ As

⁹⁸⁷ Section 3(f) of the Exchange Act requires the Commission, whenever it engages in rulemaking and is required to consider or determine whether an action is necessary or appropriate in the public interest, to consider, in addition to the protection of investors, whether the action will promote efficiency, competition, and capital formation. Additionally, section 23(a)(2) of the Exchange Act requires the Commission, when making rules under the Exchange Act, to consider the impact such rules will have on competition. Exchange Act section 23(a)(2) prohibits the Commission from adopting any rule that would impose a burden on competition not necessary or appropriate in furtherance of the purposes of the Exchange Act.

⁹⁸⁸ Investors can also execute trades on other “dark” venues that do not display quotes. But because quotes are not displayed on these venues, the investor could not be certain of the execution price or of the number of shares available. *See* Proposing Release, *supra* note 11, at 80287 (addressing trading centers that do not display protected quotes).

discussed in the Proposing Release, standard economic theory suggests that liquidity providers in a competitive market will compete to provide liquidity until the spread—i.e., their compensation for providing liquidity—is equal to the break-even point given the costs of liquidity provision.⁹⁸⁹ Absent fees, rebates, and a minimum pricing increment, this break-even point for liquidity provision represents the lowest bid-ask spread at which liquidity providers (as a whole) are willing to provide liquidity (hereinafter “economic spread”).⁹⁹⁰

2. Economics of Minimum Pricing Increments

When a market has a minimum pricing increment (hereafter “tick sizes” or just “ticks”), liquidity providers quote bid and ask prices that are discrete whole number multiples of that tick. For example, if the tick is a penny, then a liquidity provider quotes prices that are a multiple of a penny, such as \$10.00 or \$10.01, but not \$10.015. There may, however, be a liquidity provider willing to quote an ask of \$10.015 and a bid of \$10.005. Were this liquidity provider to be allowed to do so, the stock would have a spread \$0.01 (i.e., the difference between the lowest ask price and bid price). However, in the presence of the \$0.01 tick, liquidity providers will quote at

⁹⁸⁹ Proposing Release, supra note 11, at 80309 n.483. See also Jonathan Brogaard & Corey Garriott, High-Frequency Trading Competition, 54 J. FIN. & QUANTITATIVE ANALYSIS 1469 (2019) (documenting that as more high-frequency liquidity providers enter the market, spreads decrease until they converge to competitive levels).

⁹⁹⁰ Although the Proposing Release did not use the phrase “economic spread,” the release employed the same concept in multiple places. See, e.g., Proposing Release, supra note 11, at 80317 (“In a competitive market, and in the absence of rebates or other price distortions, the prevailing bid or ask price would be the feasible price equal to just worse than the price that equates liquidity supply and demand.”). In a number of places where the release employed the concept of economic spread, it arose in discussions about a stock that would trade at a given price or spread absent the tick size. See Proposing Release, supra note 11, at 80304, 80309, and 80317. The spread is composed of several elements: adverse selection, inventory risk, and processing costs. See Proposing Release, supra note 11, at 80304 n.447 and 80321. See generally, Roger D. Huang & Hans R. Stoll, The Components of the Bid-Ask Spread: A General Approach, 10 REV. FIN. STUD. 995 (Winter 1997). As explained in the Proposing Release, supra note 11, at 80304 and n.447, the spread is unlikely to ever be zero due to inventory costs, adverse selection risks, the direct costs associated with providing liquidity, and trading rules meant to prevent the locking and crossing of markets. See P.C. Kumar, Bid-Ask Spreads in U.S. Equity Markets, 43 Q. J. BUS. & ECON 85 (2004).

the best feasible ask price above \$10.015, which is \$10.02, and the best feasible bid price below \$10.005, which is \$10.00.⁹⁹¹ Consequently, the stock's quoted spread would be \$0.02 instead of \$0.01, twice as wide than it would otherwise be.

Tick sizes present an economic tradeoff. As discussed in the Proposing Release, in determining what tick size is optimal for any given stock, there is a tradeoff between price competition on one hand, and incentives for liquidity provision on the other.⁹⁹² A smaller tick allows liquidity providers to better compete on price which can lead to narrower spreads, reducing costs for investors. On the other hand, a smaller tick can also lead to pennyning. Pennyning increases adverse selection costs for slower liquidity providers by making it more likely that they trade when prices are moving in an unfavorable direction relative to their positions.⁹⁹³ To compensate for these costs, liquidity providers may post less aggressive quotes—lower bid prices and higher ask prices—resulting in a wider quoted spread and worse

⁹⁹¹ As discussed in the Proposing Release, supra note 11, at 80309 nn.483-484 and accompanying text, this assumes that stock prices are expected to revert to the next worse level. This may occur because standard economic theory suggests that in a competitive market liquidity providers will compete to provide liquidity until the spread – i.e., their compensation for providing liquidity – is equal to the break-even point for liquidity provision. See also Jonathan Brogaard & Corey Garriott, High-Frequency Trading Competition, 54 J. FIN. & QUANTITATIVE ANALYSIS 1469 (2019) (documenting that as more high-frequency liquidity providers enter the market, spreads decrease until they converge to competitive levels). The range of infeasible quoting prices narrows somewhat in the presence of rebates for liquidity providers. section VII.B.3 discusses these effects.

⁹⁹² Proposing Release, supra note 11, at 80305.

⁹⁹³ Id., at 80305 n.481 and accompanying text.